Vivekanand Education Society's Institute of Technology

V.E.S. Since 1962

Department of Computer Engineering

Group No.: 24
Date :- 11-08-2025

Synopsis (2025-26) - Sem VII

Interview IQ

Mrs. Vidya S. Zope
Asst. Professor , CMPN

Toshit Hole	Shreyas Bagwe	Jatin Hargunani	Subodh Thalkari
V.E.S.I.T	V.E.S.I.T	V.E.S.I.T	V.E.S.I.T
2022.toshit.hole@ves.ac.in	2022.shreyas.bagawe@ves.ac.in	2022.jatin.hargunani@ves.ac.in	2022.subodh.thalkari@ves.ac.in

Abstract:

InterviewIq is an innovative AI-powered mock interview platform designed to simulate highly realistic and interactive interviews across a wide range of professional domains. Utilizing advanced intelligent agents, the system dynamically generates domain-specific questions tailored to user-selected difficulty levels and real-time performance. InterviewIq incorporates multi-persona AI interviewers equipped with voice synthesis technology, enabling engaging and natural conversational flows that adapt dynamically through follow-up questions based on candidate responses. Built on a robust tech stack with a React frontend, Node.js backend, and Python-powered AI services, the platform delivers personalized interview experiences enhanced by detailed post-interview analytics and actionable performance insights. By addressing gaps in existing preparation tools, InterviewIq democratizes access to adaptive, immersive, and effective interview practice, empowering users to build confidence and improve skills in a competitive job market.

Introduction:

The landscape of modern employment, particularly within technology and specialized sectors, demands candidates not only possess strong technical competencies but also demonstrate effective communication, problem-solving under pressure, and adaptability. Conventional interview preparation methods—such as static question banks, group coaching, and self-study—often fail to capture the nuanced, dynamic nature of real interviews. These traditional approaches lack tailored feedback, realistic interviewer interactions, and the ability to simulate various interview formats or difficulty levels. Furthermore, many advanced coaching services are prohibitively expensive, limiting accessibility. Recent breakthroughs in artificial intelligence, including large language models (LLMs), speech recognition systems, and text-to-speech (TTS) synthesis, have unlocked new potential for building sophisticated virtual interview platforms. InterviewIq leverages these advancements to provide a highly adaptive, immersive, and personalized interview preparation experience that closely mirrors actual interviewing scenarios across multiple domains, making effective practice more accessible and impactful.

Problem Statement:

Current interview preparation platforms predominantly rely on static question repositories without the ability to dynamically adjust question difficulty or personalize content based on user performance trends. This results in a one-size-fits-all approach that does not cater to individual learning needs. Additionally, many platforms fail to incorporate natural conversational interactions such as adaptive follow-up questions or behavioral cues from interviewers, which are critical for realistic practice. The code execution environments offered by many services are often limited and do not replicate real-world development constraints and toolchains, reducing their effectiveness for technical interview preparation. Moreover, a lack of comprehensive analytics tools prevents candidates from gaining detailed insights into their strengths and weaknesses, hindering targeted improvement. These limitations particularly disadvantage candidates who do not have access to premium coaching, creating inequity in interview readiness and reducing overall success rates in competitive hiring processes.

Proposed Solution:

InterviewIq addresses these challenges by delivering an end-to-end AI-driven mock interview platform designed for realism, adaptability, and personalization. The system features multiple specialized AI agents that generate questions dynamically, tailored to the user's selected domain, experience level, and ongoing performance data, ensuring progressive difficulty adjustment. A diverse set of intelligent interviewer personas—each with distinct communication styles and voice profiles created through advanced speech synthesis—engage candidates in natural, lifelike conversations, including context-aware follow-up questions. The platform curates personalized learning resources and study materials aligned with identified knowledge gaps, enabling users to focus their preparation efficiently. Post-interview analytics provide in-depth performance breakdowns, highlighting strengths, weaknesses, and improvement areas, supported by recorded interview sessions that facilitate self-review and reflective learning. This comprehensive approach creates a scalable, immersive environment that closely replicates actual interview conditions and optimizes user growth.

Methodology / Block Diagram:

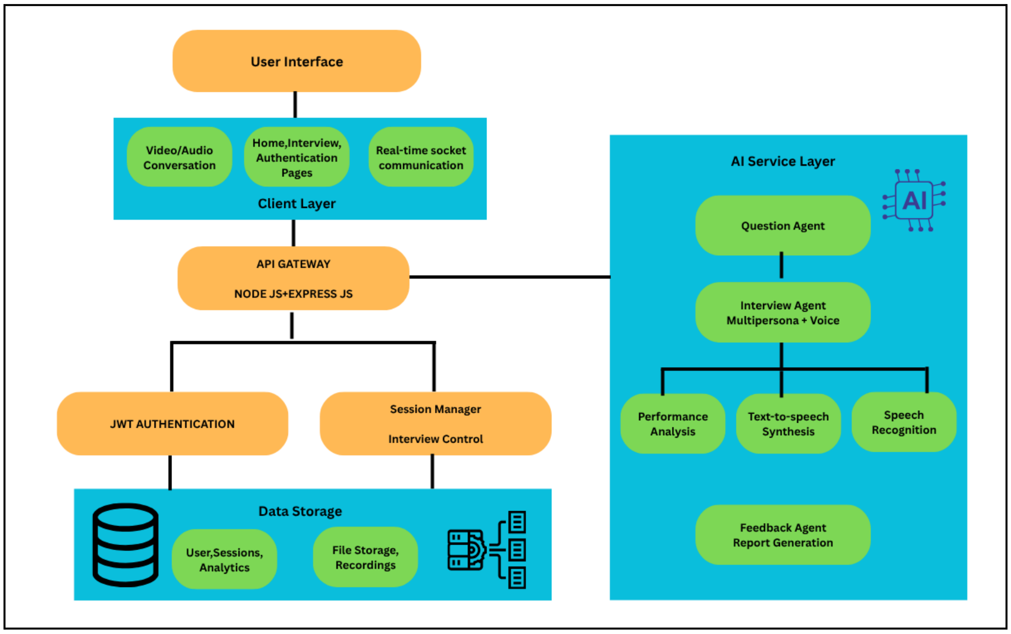
1. **User Interface Layer:**
    a. React-based frontend delivering interactive UI elements for interview setup, live sessions, and analytics dashboards.
    b. Voice input/output components using WebRTC and TTS for real-time audio communication.

2. **Backend Services:**
    a. Node.js Express server managing API requests, session control, and user data persistence through MongoDB.
    b. Socket.io for low-latency bi-directional communication enabling smooth real-time interactions.

3. **AI Services Layer:**
    a. Python microservices implementing intelligent question generation using Hugging Face Transformers and LangChain for context management.
    b. Speech recognition powered by Whisper ASR for converting user voice to text.
    c. Voice synthesis modules utilizing Coqui TTS/Bark TTS for realistic interviewer personas.

4. **Analytics & Reporting:**

   a. Data aggregation modules analyzing response accuracy, timing, speech patterns, and behavioral cues.
   b. Visualization tools providing detailed session reports and progress tracking.



Hardware , Software and tools Requirements:

- **Hardware:** Standard development laptop/PC with at least 8GB RAM, Intel i5/AMD Ryzen processor, and stable internet connection.
- **Software:** Node.js, MongoDB, Python 3.10+, Docker, Git, VS Code, Postman.
- **AI Tools/Libraries:** Hugging Face Transformers, LangChain, OpenAI API, Whisper ASR, Coqui TTS/Bark TTS.
- **Frontend:** React, Tailwind CSS, CodeMirror, WebRTC, Socket.io-client.
- **Backend:** Express.js, Mongoose, Socket.io.

Proposed Evaluation Measures:

- **Performance Metrics:**
  - Monitor and optimize AI-generated question and feedback response latency, aiming for sub-2 second delay to ensure seamless user experience.
  - Conduct stress testing to evaluate the system's ability to handle concurrent users without degradation in performance.

- **Scalability Testing:**
    - Simulate varied user loads to validate backend scalability and front-end responsiveness under real-world conditions.
- **Technical Validation:**
    - Benchmark against existing interview platforms to demonstrate superior personalization, adaptive questioning, and detailed analytics.

Conclusion:

InterviewIq stands as a pioneering AI-driven platform that revolutionizes interview preparation by integrating multi-agent intelligence, dynamic question generation, and natural conversational AI. Its holistic approach combines personalized learning pathways, immersive interviewer personas, and deep analytics to comprehensively develop candidate skills, confidence, and readiness. By leveraging state-of-the-art AI technologies and scalable architecture, InterviewIq democratizes access to quality interview coaching regardless of geographic or economic barriers. The anticipated benefits include measurable improvements in interview outcomes, enhanced user motivation, and a more equitable hiring preparation landscape. Future iterations aim to broaden domain coverage, incorporate advanced behavioral and emotional analysis, and establish seamless integrations with recruitment systems, thereby closing the loop from preparation to placement.

References:

- Chen, L., Wang, S., & Liu, H. (2023). *Artificial Intelligence in Education: Applications and Methodologies*. 4th Edition. Academic Press, Boston, MA.
- Johnson, M. K., Thompson, R. A., & Davis, P. (2022). *Web Development with Modern JavaScript Frameworks*. 2nd Edition. O'Reilly Media, Sebastopol, CA.
- Rodriguez, A., Kumar, V., & Singh, P. (2023). *Machine Learning Systems Design: Scalable AI Applications*. 3rd Edition. MIT Press, Cambridge, MA.
- Wilson, J. D., Brown, K. L., & Taylor, S. M. (2022). *Secure Software Architecture: Containerization and Microservices*. 1st Edition. Addison-Wesley Professional, Boston, MA.
- Zhang, Y., Anderson, C., & Williams, D. (2023). *Speech Recognition and Natural Language Processing: Modern Approaches*. 2nd Edition. Springer Nature, New York, NY.
- OpenAI Development Team. (2024). *OpenAI API Documentation: Audio and Language Models*. OpenAI Inc., San Francisco, CA.
- Mozilla Foundation. (2023). *WebRTC 1.0: Real-Time Communication Between Browsers*. W3C Recommendation, World Wide Web Consortium.